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Original Article

Open Versus Robot-assisted Kidney Transplantation in Adults with End-stage Kidney Disease (ORAKTx): A Randomised Open-label Trial

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Article info

Article history:

Accepted April 18, 2026

Associate Editor:

Gianluca Giannarini

Statistical Editor:

Andrew Vickers

Keywords:

End-stage kidney disease
Kidney transplantation
Randomised controlled trial
Robot-assisted kidney transplantation
Robotic surgery
Urology

Abstract

Background and objective: We investigated whether robot-assisted kidney transplantation (KTx) could reduce short-term surgical complications compared to open KTx.

Methods: We conducted a randomised, parallel, open-label trial at the Copenhagen University Hospital – Rigshospitalet, Denmark. Adult patients with end-stage kidney disease and no contraindication to robotic surgery, scheduled to undergo living or deceased donor KTx, were randomly assigned to receive either open or robot-assisted surgery. Primary outcomes were the 30-d post-transplant rates of vascular and major surgical complications. All participants who underwent transplantation were analysed according to assigned intervention.

Key findings and limitations: A total of 54 participants were assigned to robot-assisted KTx and 53 were assigned to open KTx. Fifty-three participants in each arm underwent transplantation and completed follow-up. Major surgical complications occurred in 13 patients (25%) in the open arm and in seven patients (13%) in the robotic arm (risk difference: –11%, 95% confidence interval [CI]: –26 to 3.4, $p = 0.2$). Vascular complications occurred in 10 patients (19%) in the open arm and in five patients (9.4%) in the robotic arm (risk difference: –9.4%, 95% CI: –23 to 3.7, $p = 0.3$). Limitations include single-centre design, lack of blinding, and a modest sample size.

Conclusions and clinical implications: ORAKTx is the first randomised trial to compare robot-assisted KTx with open KTx surgery. Although the primary endpoint was not met, the observed reduction in complications suggests a potentially meaningful decrease in postoperative morbidity with robot-assisted surgery.

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ADVANCING PRACTICE

What does this study add?

ORAKTx is the first randomised trial to compare robot-assisted kidney transplantation (KTx) to open KTx, providing a comprehensive and systematic overview of short-term surgical morbidity following both procedures. The trial is marked by wide patient selection criteria, including deceased donor transplantations, unlike previous studies. The results did not demonstrate superiority of robot-assisted KTx over open KTx, but we observed an approximately 50% reduction in both vascular and major surgical complications, indicating that minimally invasive surgery could play an important role for the general population undergoing KTx.

Clinical Relevance

This randomized trial provides the first high-level evidence comparing robot-assisted versus open kidney transplantation across a broad, real-world population, addressing a critical gap left by prior observational studies. Although superiority was not statistically demonstrated, the consistent reduction (of approximately 10%) in major and vascular complications suggests a clinically meaningful signal toward reduced perioperative morbidity with the robotic approach. For urologists and transplant surgeons, these findings support cautious adoption in selected settings while underscoring the need for larger multicentre trials to define patient selection, cost-effectiveness, and the true impact on functional outcomes. Associate Editor: Gianluca Giannarini, MD.

Patient Summary

This study examined whether the use of robot-assisted laparoscopy could improve surgical outcomes for patients undergoing KTx. We noted differences in the pattern of complications between the two types of surgery, favouring the use of robotics. However, due to the relatively small number of patients and complications in this study, we could not definitively conclude that robot-assisted surgery is superior to open surgery in KTx.

1. Introduction

Recent decades have seen several advances in immunosuppressive therapy, donor selection, and organ preservation in kidney transplantation (KTx) for patients with end-stage kidney disease (ESKD) [1,2]. Despite these advances, the surgical aspects of KTx have stayed largely unchanged, and the current standard of care in KTx remains open surgery [3,4]. Postoperative complications following open KTx remain important contributors to early morbidity and can compromise renal outcomes [5–7]. To improve surgical outcomes, robot-assisted KTx has been used as a minimally invasive alternative to open KTx, and is being increasingly adopted by the urological and global transplant community [8,9].

Several systematic reviews comparing outcomes between minimally invasive, robot-assisted KTx and open KTx have been performed, but none have included randomised trials [8,10–17]. Meta-analyses from these reviews suggest that robot-assisted KTx is associated with improved recovery and a lower risk of wound-related and major surgical complications. Importantly, the included studies were largely performed in a living-donor setting and thus only reflect a minor proportion of KTx recipients [18]. This discrepancy between the lack of randomised trials and the selected cohorts in which robot-assisted KTx has been performed limits the generalisability of its observed advantages over open surgery concerning surgical outcomes.

With this trial, we aimed to assess the hypothesis that robot-assisted KTx could offer substantial reductions in

short-term major and vascular complications compared to open KTx. Prior to trial initiation, we conducted a pilot study to ensure the feasibility of robot-assisted KTx at our institution, confirming a rationale for selecting these complications as trial outcomes [19]. Additionally, we assessed recovery, urological complications, and graft function.

2. Materials and methods**2.1. Study design**

We conducted this randomised, open-label, parallel group trial at the Copenhagen University Hospital – Rigshospitalet, a tertiary referral centre for KTx. The study was ethically approved by The Committees on Health Research Ethics in the Capital Region of Denmark (H-22065569), registered with clinicaltrials.gov (NCT05730257) on Feb 7, 2023, and the protocol was published [20]. Primary and key secondary outcomes are presented in this study, while long-term secondary outcomes, including patient-reported quality of life, will be presented in later reports.

2.2. Participants

Adult patients (≥ 18 yr) scheduled to undergo single KTx at Rigshospitalet were eligible to participate if surgeons deemed them suitable to undergo robot-assisted KTx. This included abdominal anatomy that allowed transperitoneal access to, and placement of, the kidney graft in the iliac fossa, no evidence of severe calcifications of the iliac vessels, and no comorbidities contraindicating the Trendelenburg

position ([Appendix p 4](#)). For recipients of deceased donor KTx, where the need for surgery presents acutely, inclusion required availability of both the robotic operating room and robotic KTx team. All participants gave written, informed consent.

2.3. Randomisation and masking

Participants were randomly assigned in a 1:1 ratio to receive open or robot-assisted KTx using permuted blocks of variable size (4 and 6), without stratification. The allocation sequence was computer-generated by an independent researcher and implemented in the secure, access-restricted randomisation module of the research electronic data capture (REDCap) software prior to study initiation, with block sizes concealed from investigators and surgeons [[20,21](#)]. Recipients were enrolled by surgeons and trial investigators who coordinated logistics with the operating room. The investigator performing randomisation had no access to the sequence and could not predict or influence upcoming assignments. Randomisation was performed after eligibility had been confirmed and the participant was irrevocably registered in REDCap ([Appendix p 4](#)). Once randomised, the assigned allocation was automatically revealed and locked to the record, precluding any modification. Variable block sizes and sequence concealment ensured that prior allocations could not be used to subvert assignment. All clinical decisions after randomisation were at the discretion of treating clinicians. The trial was open-label; however, the primary data analyst was blinded to group allocation.

2.4. Procedures

Both surgical approaches used an end-to-side anastomosis between the kidney graft vessels and the recipient's external iliac vessels and an extravesical approach to the ureteroneocystostomy, including a ureteric stent and a bladder catheter ([Appendix p 6](#)).

For open KTx, the procedure is entirely retroperitoneal, and access was made using a Gibson incision in the iliac fossa. For robot-assisted KTx, the procedure is transperitoneal, applying a Pfannenstiel incision to introduce the kidney graft, which is continuously cooled.

Anaesthesiology and immunosuppressive therapy followed local protocols ([Appendix pp 6–8](#)).

Surgeon participation was restricted to experienced surgeons, but no quantitative thresholds were set. All senior consultant surgeons who performed open KTx independently, prior to trial enrolment, were eligible to participate in the open arm. Two surgeons were eligible to administer robot-assisted KTx. Both were experienced in KTx, open and robotic surgery (specific caseloads available in the [Appendix p 4](#)). Prior to trial enrolment, a pilot study of 16 cases was performed by the two surgeons, who were assessed as having adequately attained the learning curve for robot-assisted KTx based on operative times and complication rates [[19](#)].

2.5. Outcomes

The trial measured two primary outcomes, assessed at 30-d post-transplant: the proportion of recipients with a major surgical complication and the proportion of recipients with a graft-related vascular complication. A complication was defined according to the Clavien-Dindo classification, and major complications were defined as any complication graded $\geq 3a$ [[20,22](#)]. Vascular complications were bleeding requiring reoperation, symptomatic hematomas, vascular thrombosis, or arterial stenosis ([Appendix p 9](#)). The adjudication of a primary outcome required agreement between two assessors.

Secondary outcomes reported in this study were recovery, kidney function, and urological complications at 30-d post-transplant, as defined in the protocol ([Appendix p 9](#)) [[20](#)]. The primary outcomes and urological complications were additionally explored at 90-d post-transplant. All outcomes, as well as severe adverse events (SAEs), were captured through the recipients' electronic health records (EHR) and include out-patient information.

2.6. Statistical analysis

The trial was designed to demonstrate a 15% decrease in the proportion of recipients with vascular complications (from 17% in the open group to 2% in the robotic group) and a 20% decrease in the proportion of recipients with major complications (from 23% in the open group to 3% in the robotic group). The expected 17% and 20% frequencies in the open arm were based on retrospective data from our institution, with symptomatic hematomas and Clavien-Dindo grade 3 complications driving both outcomes [[23](#)]. With no estimates available through meta-analyses, the size of the reduction offered by robotics was chosen through the theoretical considerations that robotics could reduce intervention-requiring peri-graft collections through transperitoneal absorption (hematomas and Clavien-Dindo grade 3a complications), produce superior anastomotic results through technical advantages on both the vascular and ureteral sides (reducing hematomas and Clavien-Dindo grade 3a and 3b complications), reduce surgical stress through minimal invasiveness (Clavien-Dindo grade 4 complications), and our practical experience in the pilot study that demonstrated a 0% rate of vascular and major complications at 90-d post-transplant. To obtain 80% power with an α risk of 5% and assuming 10% attrition, we planned to randomise 106 participants ([Appendix p 3](#)).

Participants were analysed according to the assigned intervention if they underwent KTx (intention-to-treat). Descriptive statistics were computed for the baseline characteristics of the two groups. Continuous variables were described as median with interquartile range, and categorical data as absolute and relative frequencies.

Rates of binary outcomes were reported as counts with percentages. Between-treatment comparisons of binary outcomes were assessed by chi-square test, accompanied by absolute risk differences, risk ratio, and 95% confidence

interval (CI) estimates for the primary outcomes. Ordinal categorical outcomes were compared by chi-square test. Continuous outcomes were assessed by Wilcoxon rank-sum test. The primary outcomes were explored in a stratified analysis for living and deceased donors. Reported *p* values are two-sided and unadjusted for multiple testing, with values <0.05 considered as statistically significant. Missing data were minimal and were presented descriptively.

3. Results

Participants were recruited between May 8, 2023 and Sep 24, 2025. A total of 231 intended KTx recipients were assessed for eligibility, of whom 107 were recruited (Fig. 1). Fifty-three participants were assigned to receive open KTx and 54 were assigned to receive robot-assisted KTx. Of these, 106 underwent KTx and were assessed for the primary endpoint. The most common exclusion criteria were unavailability of the robotic team (60 [48%]) and abdominal anatomy that did not allow access to, and placement of, the graft in the iliac fossa (20 [16%], Appendix p 5).

Baseline characteristics were overall well-balanced between the two groups, with 59 recipients (56%) receiving a graft from a deceased donor (Tables 1 and 2). All characteristics of recipients, donors, grafts, and immunology are available in the Appendix (pp 10–12). There were no intra-operative major adverse events in the robotic arm and no conversions to open surgery (Table 3). All intraoperative characteristics are available in the Appendix (p 13).

At 30-d post-transplant, major surgical complications occurred in 13 recipients (25%) in the open arm compared

Table 1 – Baseline characteristics of recipients

	Open arm, <i>n</i> = 53	Robotic arm, <i>n</i> = 53
Age, yr	47 (37–59)	51 (36–59)
Body mass index, kg/m ²	25 (22–28)	26 (24–30)
Female	24 (45)	16 (30)
Male	29 (55)	37 (70)
Charlson comorbidity index		
0–2	46 (87)	46 (87)
≥3	7 (13)	7 (13)
Previous abdominal surgery	16 (30)	19 (36)
Previous kidney transplantation	5 (9.4)	5 (9.4)
Cause of kidney disease		
APKD	4 (7.5)	9 (17)
Chronic pyelonephritis, interstitial nephritis, and reflux nephropathy	3 (5.7)	4 (7.5)
Diabetic nephropathy	3 (5.7)	2 (3.8)
Glomerulonephritis	21 (40)	15 (28)
Hypertension and arteriosclerosis	3 (5.7)	4 (7.5)
Systemic disease	3 (5.7)	2 (3.8)
Unidentified	11 (21)	10 (19)
Other	5 (9.4)	7 (13)
Previous dialysis ^a	38 (72)	39 (74)
Haemodialysis	28 (53)	29 (55)
Peritoneal dialysis	9 (17)	10 (19)
Serum creatinine, μmol/l	692 (498–841)	667 (502–889)

APKD = adult polycystic kidney disease; IQR = interquartile range. Data are *n* (%) or median (IQR). The Charlson comorbidity index is adjusted for chronic kidney disease and age. One patient in the open arm received hybrid dialysis. There are no missing data.

to seven recipients (13%) in the robotic arm. Although the proportion is lower for patients in the robotic arm, it was not statistically significant (risk difference: –11%, 95% CI: –26 to 3.4; risk ratio: 0.54, 95% CI: 0.23–1.24, *p* = 0.2).

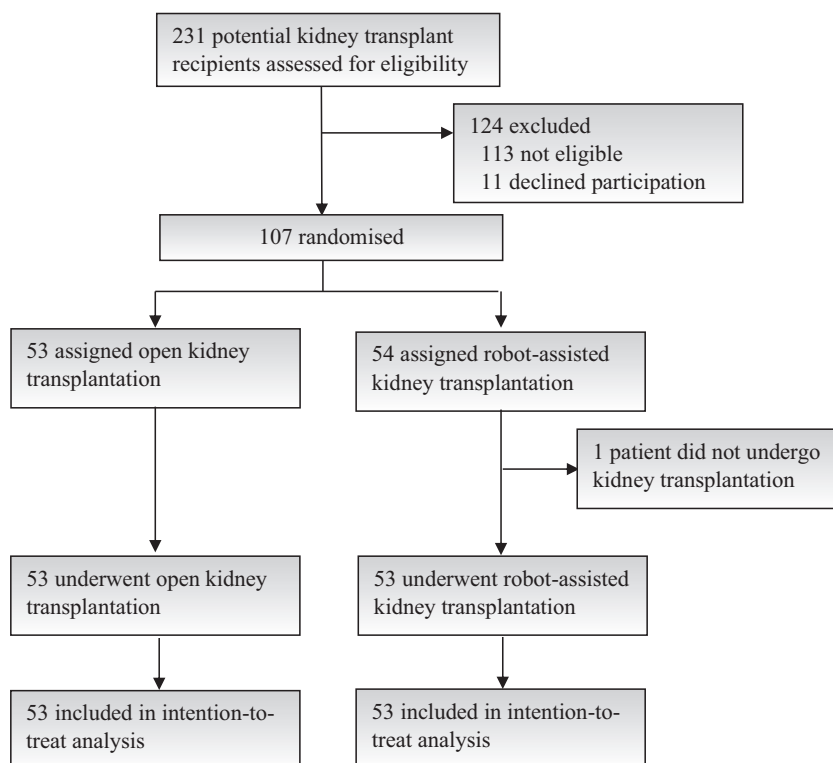


Fig. 1 – Trial profile.

Table 2 – Baseline characteristics of donors, grafts, and immunology

	Open arm, n = 53	Robotic arm, n = 53
Age, yr	53 (46–65)	61 (45–71)
Female	25 (47)	30 (57)
Male	28 (53)	23 (43)
Living donor	25 (47)	22 (42)
Deceased donor	28 (53)	31 (59)
DBD	21 (75)	22 (71)
DCD	7 (25)	9 (29)
Left graft	32 (60)	29 (55)
Single artery graft	43 (81)	41 (77)
Single vein graft	52 (98)	52 (98)
ABO-compatible KTx	51 (96)	51 (96)
Standard immunosuppression	47 (89)	45 (85)
CIT living donor KTx, min	162 (90–210)	116 (99–123)
CIT deceased donor KTx, h	14 (7–18)	14 (8–21)

ABO = ABO blood grouping system; CIT = cold ischemia time; DBD = donation after brain death; DCD = donation after circulatory death; IQR = interquartile range; KTx = kidney transplantation. Data are n (%) or median (IQR). There are no missing data.

Graft-related vascular complications at 30-d post-transplant occurred in 10 recipients (19%) in the open arm and five recipients (9.4%) in the robotic arm. The lower proportion for recipients in the robotic arm was likewise not statistically significant (risk difference: -9.4% , 95% CI: -23 to 3.7 ; risk ratio: 0.50 , 95% CI: 0.18 – 1.36 , $p = 0.3$). There were no substantial differences in major, vascular, or urological complications at 90-d post-transplant (Appendix p 18).

Five recipients (9.4%) in the open arm experienced urological complications compared to four recipients (7.5%) in the robotic arm (risk difference: -1.9% , 95% CI: -8.7 to 13 , $p = 1$). Two recipients (3.8%) had primary non-function of the kidney graft after open KTx whereas one recipient (1.9%) in the robotic arm underwent graftectomy due to venous thrombosis (risk difference: -1.9% , 95% CI: -8.2 to 4.4 , $p = 1$). No differences between groups were observed for kidney function or recovery, except that recipients in

Table 3 – Intraoperative characteristics

	Open arm, n = 53	Robotic arm, n = 53	Absolute/estimated difference	p value
Total operative time, min	131 (114–150)	184 (154–198)	49 (36–63)	<0.001
Blood loss, ml	240 (100–400)	150 (50–350)	–70 (–140 to 0)	0.069
Drain placement	8 (15)	16 (30)	15 (–0.6 to 31)	0.10
Conversion to open surgery	–	0	–	–
Major adverse events	2 (3.8)	0	–3.8 (–1.4 to 8.9)	0.5

Data are n (%), median (interquartile range [IQR]), absolute difference % (95% confidence interval [CI]), or Hodges-Lehmann estimated difference between groups (95% CI). Both cases of intraoperative adverse events were major bleeding (>1l).

Table 4 – Surgical and functional outcomes at 30-d posttransplant

	Open arm, n = 53	Robotic arm, n = 53	Risk difference/estimated difference	p value
Primary outcomes				
Major complications	13 (25)	7 (13)	–11 (–26 to 3.4)	0.2
Clavien-Dindo grades, highest given				
3a	5 (9.4)	0		
3b	5 (9.4)	6 (11)		
4a	3 (5.7)	1 (1.9)		
Vascular complications	10 (19)	5 (9.4)	–9.4 (–23 to 3.7)	0.3
Bleeding requiring reoperation	3 (5.7)	2 (3.8)		
Symptomatic hematoma	7 (13.2)	1 (1.9)		
Venous thrombosis	0 (0)	2 (3.8)		
Arterial stenosis	1 (1.9)	0		
Secondary outcomes				
Length of stay, d	7 (6–12)	8 (7–12)	0 (–1 to 1)	0.6
Days alive and out of hospital	22 (17–23)	22 (18–23)	0 (–1 to 1)	0.8
Opioid use, MME/d	20 (14–26)	29 (21–43)	8.5 (3.6–14)	0.001
Patients requiring transfusion	17 (32)	17 (32)	0 (–18 to 18)	1
Erythrocytes, units	3 (2–5)	2 (1–4)	–1 (–2 to 1)	0.3
Urological complications	5 (9.4)	4 (7.5)	–1.9 (–13 to 8.7)	1
Extravasation of urine	0	1 (1.9)		
Hydronephrosis	2 (3.8)	3 (5.7)		
Lymphocele	4 (7.5)	0		
Delayed graft function	10 (19)	9 (17)	–1.9 (–13 to 17)	1
eGFR, ml/min/1.73 m ²				0.8
≥90	2 (3.8)	2 (3.8)	–	
60–89	11 (21)	10 (19)		
45–59	12 (23)	16 (30)		
30–44	15 (28)	13 (25)		
15–29	9 (18)	11 (21)		
<15	4 (7.5)	1 (1.9)		
Serum creatinine, μmol/l	148 (118–174)	151 (124–189)	4 (–16 to 23)	0.7

CI = confidence interval; eGFR = estimated glomerular filtration rate; IQR = interquartile range; MME = morphine milligram equivalents.

Data are n (%), median (IQR), absolute risk difference % (95% CI), or Hodges-Lehmann estimated difference between groups (95% CI). No patients experienced arterial thrombosis or grade 4b or 5 complications. There are no missing data.

the robotic arm received significantly more opioids (Table 4).

In the stratified analysis of major complications for living donor KTx, the rate was 24% in the open arm and 9.1% in the robotic arm (risk difference: -15% , 95% CI: -36 to 5.7 ; risk ratio: 0.38 , 95% CI: 0.09 – 1.69 , $p = 0.3$). The deceased donor analysis demonstrated a major surgical complication rate in the open arm of 25% compared to the 16% in the robotic arm (risk difference: -8.9% , 95% CI: -30 to 12 ; risk ratio: 0.65 , 95% CI: 0.23 – 1.80 , $p = 0.6$).

The stratified analysis of vascular complications in living donor KTx demonstrated a 30-d rate of 16% in the open arm and 0% in the robotic arm (risk difference: -16% , 95% CI: -30 to -1.6 , $p = 0.2$). For deceased donor KTx, the rate of vascular complications was 21% in the open arm and 16% in the robotic arm (risk difference: -5.3% , 95% CI: -25 to 15 ; risk ratio: 0.75 , 95% CI: 0.26 – 2.20 , $p = 0.9$).

There were no differences among patients with SAEs between the groups. A complete list of SAEs and complications, including their assigned Clavien-Dindo grade, is available in the Appendix (pp 14–17).

4. Discussion

In this first of its kind randomised trial comparing robot-assisted and open KTx, we demonstrated an absolute risk reduction of approximately 10% in clinically important complications with robot-assisted surgery; although the reduction was not statistically different as the trial was powered to detect a larger reduction. There were no differences in secondary outcomes, except higher rate of opioid use, for patients in the robotic arm.

The ORAKTx trial represents a pivotal advance in surgical research in KTx, providing the first randomised evidence to challenge open surgery as the standard approach for reducing clinically important complications. A major innovation of this trial is its pragmatic inclusion of all-comers, including patients undergoing emergent transplantation, thereby evaluating minimally invasive surgery across the full clinical spectrum.

Surgical innovation in KTx has not followed its medical counterparts, an inertia that likely reflects the high stakes setting of KTx: emergency surgery, medically complex recipients, limiting ischaemia times, and the logistical complexities of organ allocation combined with the expectation of success. Robot-assisted surgery has transformed urological practice over the past two decades, with the 2022 iROC trial highlighting the potential advantages of minimally invasive techniques in high-risk surgical populations [24]. These findings motivated our interest in evaluating whether similar benefits could be achieved in patients with ESKD undergoing KTx—a group characterised by high susceptibility to perioperative morbidity. Additionally, evidence for robot-assisted surgery in emergency settings is scarce, as reflected in the 2021 position statement from the World Society of Emergency Surgery, which emphasised the need for randomised data to inform practice in acute surgery [25].

Surgical complications in open KTx are common, but the transplant literature has been slow to adapt standardised

classification systems, hindering comparison of surgical outcomes. Studies that apply a validated system report similar major complication rates to those observed in our open KTx cohort [26].

The observed rate of major complications for living donors and length of stay in the robotic arm is consistent with observational studies [9]. A reduction in major complications could theoretically be attributed to a reduced surgical stress response associated with a minimally invasive approach [27]. In contrast to previous studies, we did not observe a reduction in lymphoceles, but our definition was not limited to lymphoceles requiring intervention [17]. A reduction in fluid collections surrounding the graft can be explained by the transperitoneal approach in robotics, allowing for the intra-abdominal space to absorb fluid surrounding the graft. This is also reflected in the outcome of major complications where Clavien-Dindo grade 3a complications (postoperative drain placement) were insignificant in the robotic arm. Notably, the rate of vascular complications in the open group was driven by symptomatic hematomas, whereas the robotic group had two cases of venous thrombosis. Graft thrombosis directly compromises graft survival and may not be of equal importance to patients as symptomatic hematomas [28]. Importantly, no differences in graft function were observed. Though baseline characteristics were overall well-balanced, the open arm had a larger proportion of female recipients, which has been associated with an increased risk of vascular complications, but conversely, the proportion of deceased donor KTx was higher in robotic group and has likewise been associated with an increased risk of bleeding [23,29]. Furthermore, a substantial proportion of patients were excluded from participation due to abdominal anatomy that was deemed unfit for robotic surgery, such as a history of major intra-abdominal surgery.

We did not observe a pattern indicating reduced morbidity in our secondary outcomes, but KTx recipients often require readmission for reasons unrelated to surgery, such as rejection [19]. Recipients in the robotic arm received more opioids postoperatively in contrast to observational studies [30]. The transperitoneal approach, including use of graft cooling, may cause bowel paralysis and pain leading to increased use of analgesics. We did not alter concomitant care prior to trial initiation, nor did we train nephrological nurses in the procedure-specific differences, which may have affected the outcome. Patient education on expected pain patterns following transperitoneal surgery may hold potential for improving recovery and for reducing opioid use.

The strengths of this trial are the inclusion of deceased donor KTx, experience of the participating surgeons, and standardisation of the delivered surgical interventions. We applied a thorough and systematic approach to the complication reporting, with rates comparable to those observed in our retrospective review, indicating robustness of the outcome assessment.

Limitations to this study are generalisability due to the single-centre design and risk of bias due to the lack of masking. Experience in both robotic surgery and open KTx are crucial to extrapolate results of this trial to other settings

but are also characteristic for urological centres that undertake robot-assisted KTx. The inclusion of emergency KTx did not allow masking of clinicians and patients, and outcome assessment through EHRs was deemed essential for complete assessment of complications. We believe detection bias is limited due to the prespecified definitions of the primary outcomes and two person-agreement to confirm whether an event constituted a primary endpoint. The trial sample size is modest, based on a hypothesis that robot-assisted KTx could offer a large reduction in complications, with two primary outcomes analysed nonhierarchically. The two outcomes were chosen from a hypothesis that robotic surgery reduced certain complications, coupled with a concern it may be associated with other important complications that could not be prespecified.

At the time of protocol development, limited robust data on major and vascular complication rates following both procedures were available, and thus the estimates for open KTx were based on a retrospective review from our institution and the estimates for robot-assisted KTx were based on our experience from the pilot study [17,19]. However, the pilot study was limited to a small cohort of living donor robot-assisted KTx, resulting in possible overestimation of the reduction that could be achieved by using robotics. Though we assessed participating robotic surgeons to have curbed the learning curve for robot-assisted KTx based on pretrial operative times and complications, there is a risk that this trial was conducted during a potential learning curve phase, particularly for deceased donor KTx. Notably, for living donor robot-assisted KTx, the expected reduction of 15% in vascular complications was attained as predicted in the power calculation.

This trial was designed from a pragmatic perspective, and thus we restricted participation to experienced surgeons but not exclusively experts. We aspired to demonstrate large reductions associated with robotics from a patient and health care perspective, which seemed attainable based on our previous experience and considered trial feasibility, in the absence of prior studies.

Lastly, the optimal candidate for robot-assisted KTx remains uncertain, and whether this is a procedure for all-comers or it should be restricted to patients with obesity can be continuously debated [31].

5. Conclusions

Although the prespecified primary endpoint was not met, ORAKTx demonstrated an absolute reduction of approximately 10% in major and vascular complications, which could be clinically relevant. Though these results have no immediate implications for clinical practice, they reinforce the rationale for larger multicentre confirmatory trials of robot-assisted KTx, for which ORAKTx now provides the critical evidence base for future sample-size calculations and outcome selection. Based on ORAKTx data, a trial targeting an approximately 10% reduction in complications in the robotic arm would require around 450 participants, but it could incorporate cost-benefit analyses to determine if this level of postoperative morbidity reduction can economically justify robot-assisted KTx in the future.

Author contributions: Milla Ortvad had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Study concept and design: Røder, Dagnæs-Hansen, Rohrsted, Sørensen, Kistorp, Ortvad, Stroomberg.

Acquisition of data: Dagnæs-Hansen, Røder, Ortvad.

Analysis and interpretation of data: Røder, Dagnæs-Hansen, Ortvad, Stroomberg, Rohrsted, Kistorp, Sørensen.

Drafting of the manuscript: Røder, Ortvad.

Critical revision of the manuscript for important intellectual content: Røder, Sørensen, Rohrsted, Karas, Vejlsturp, Ruhnau, Kistorp, Dagnæs-Hansen, Stroomberg, Ortvad.

Statistical analysis: Ortvad, Stroomberg.

Obtaining funding: Røder, Dagnæs-Hansen, Ortvad.

Administrative, technical, or material support: Røder, Ortvad, Dagnæs-Hansen, Ruhnau, Vejlsturp, Kistorp, Karas, Rohrsted, Sørensen.

Supervision: Røder, Dagnæs-Hansen, Ruhnau, Sørensen.

Other (specify): None.

Financial disclosures: Milla Ortvad certifies that all conflicts of interest, including specific financial interests and relationships and affiliations relevant to the subject matter or materials discussed in the manuscript (eg, employment/affiliation, grants or funding, consultancies, honoraria, stock ownership or options, expert testimony, royalties, or patents filed, received, or pending), are the following: None.

Funding/Support and role of the sponsor: This trial was financially supported by the Novo Nordisk Foundation (NNF21OC0072224), Nyre-foreningens Forskningsfond, and Rigshospitalet Forskningsudvalg. The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Acknowledgements: We thank Dr. Mie Ynddal and Dr. Maja Vejlgård for their time and assistance in project coordination and randomisation when Ortvad was unavailable. We are grateful to all the study patients, and recognise the valuable contributions of surgeons and medical staff at the three participating departments at Rigshospitalet without whom this trial would not have been possible.

Data sharing: Data will not be made available, but can be provided by the corresponding author after authorization from the Danish Data Protection Agency.

Supplementary data

Supplementary data to this article outlining study design, statistical framework, eligibility criteria, operative protocols, and detailed perioperative and postoperative outcomes—including baseline characteristics, intraoperative data, adverse events, and complication grading can be found online at <https://doi.org/10.1016/j.eururo.2026.04.022>.

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